Lecture 03.08: Multiplication Table of N

📌 Executive Summary

Mathematical sequence generation, arithmetic progression, formatted tabular alignment, and range iteration.

🔬 Theoretical Foundations & Principles

1. Arithmetic Progression Definition

The multiplication table of integer N for multipliers $i \in \{1, 2, \ldots, K\}$ forms an arithmetic progression with common difference $d = N$:

> $$a_i = N \cdot i, \quad a_{i+1} - a_i = N$$

2. String Format Alignment

Use Python f-string alignment specifiers (`{N:2d} x {i:2d} = {N*i:3d}`) to generate clean columnar alignment.

💻 Production Implementation & Code

```
def print_multiplication_table(n: int, max_multiplier: int = 10) -> None:
    print(f"--- Multiplication Table for N = {n} ---")
    for i in range(1, max_multiplier + 1):
        product = n * i
        print(f"{n:>3d}  x  {i:>2d}  =  {product:>4d}")

print_multiplication_table(7, 10)
```

⚙️ Algorithmic Complexity & Systems Analysis

- **Time Complexity:** Detailed in respective algorithmic operations.

- **Auxiliary Space:** Memory bounds and CPython frame management.